Annexes

A Governance Layer for AI Development

Technical Product Dossier

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Product and Reviewer Guide

Understand, challenge and verify

Start from a preserved audit

Open boveda.dev and sign in using the demonstration access supplied with the dossier. Select an audit from the library. The repository link identifies the source project; the displayed record describes the evidence retained for that audit, not necessarily the current GitHub branch.

Source: Bóveda Alpha 2.0.1; preserved product implementation and release records

Choose a supervisory question

View	Reviewer action	Evidence to inspect
Overview	Establish purpose, scope and the claim boundary.	Read Data Shape, model-specific performance and Audit Confidence together. Confidence describes the audit basis, not model quality.
Evidence	Identify what material supports the project.	Inspect sources, data context, retained previews and field-level evidence references. A preview is optional and bounded.
Construction	Follow preparation, representation and learning.	Switch between recorded routes and models. Do not infer that similarly numbered tabs establish a cross-route relationship.
Signals	Review predefined conditions that deserve attention.	Open the check, outcome, explanation and supporting evidence. Absence of a SIGNAL is not automatically CLEAR.
Visuals	Inspect additional project material.	Distinguish contextual cards, related collections, computed comparisons and source-only thumbnails.

Source: Bóveda Alpha 2.0.1; product viewer and hosted-demo documentation

A repeatable review sequence

1. Record the audit identity and repository. 2. Read its purpose and boundaries. 3. Select one performance claim and its evaluation context. 4. Inspect a Signal or unresolved field. 5. Follow the evidence to its retained location. 6. Download the canonical report and record any unresolved review question.

Source: Bóveda Alpha 2.0.1; preserved product implementation and release records

Mobile navigation

When the viewport cannot fit all five views, the remaining destinations are available in the three-dot menu. The active state identifies the selected section; the content and evidence are unchanged.

Source: Bóveda Alpha 2.0.1; preserved product implementation and release records

Product and Reviewer Guide

What the interface establishes

Keep three kinds of status separate

Status family	What it means
Field status	Documented: directly supported. Interpreted: contextual reconstruction. Derived: a supported computation. Unresolved: the available basis does not establish the field. Inspect the attached evidence and scope.
Signal outcome	SIGNAL, CLEAR, INSUFFICIENT EVIDENCE and NOT APPLICABLE answer a particular deterministic rule. These are not grades for the whole project.
Publication / execution	A completed execution can remain limited or non-publishable. Availability in the demonstration is not a certification of the source project.

Source: Bóveda Alpha 2.0.1; product viewer and hosted-demo documentation

Visuals: different evidentiary roles

Material	Permitted interpretation
Project-authored contextual visual	An original retained figure linked to relevant project context. Its presence is not independent verification of its scientific result.
Collection of related visuals	Several figures presented together when a relationship is supported. Review source locations and any caveat before comparing panels.
Bóveda-computed visual	A new representation of already established values. It does not rerun models, recover missing measurements or create new empirical evidence.
Other source images	Retained PNG/JPEG assets, including supported notebook or HTML outputs, shown with filenames and locations. No additional audit interpretation is inferred.

Source: Bóveda Alpha 2.0.1; product viewer and hosted-demo documentation

Boundaries

The gallery removes exact duplicates and loads thumbnails progressively. It is not an exhaustive inventory of every visual format: unsupported formats, oversized files and unresolved assets may be absent. No visual material is a normal state and does not degrade an audit.

Source: Bóveda Alpha 2.0.1; preserved product implementation and release records

Hosted and local operation

The hosted demonstration is read-only: new audits, reanalysis and destructive operations are disabled. The local application can create new audits and incur API expenditure. Browsing a preserved audit does not launch a new analytical run. Access and operating requirements are documented in Annex D.

Source: Bóveda Alpha 2.0.1; preserved product implementation and release records

Validation Corpus and Evidence

Scope, units of analysis, and coding rules

Two denominators

The research corpus contains 23 unique public GitHub repositories (R1–R23), each represented once at a pinned commit. The displayed product cohort contains 23 audit outputs over 21 unique repositories because NMR and SubwayScience each have an additional Fresh rerun; R10 and R20 are absent from that display cohort. This reconciliation retains 25 case records: the 23 research attempts and two Fresh reruns. It is not a count of every development execution. Repository diversity and audit executions are reported separately.

Source: VALIDATION_CORPUS.md; REVIEW.md; manifest.json

Operational definitions

Field	Meaning	Does not mean
Repository unit	One pinned GitHub repository in R1–R23	One successful or independent validation
Audit case	One Bóveda execution/output identity	One unique repository
Fresh	A separate rerun of an already represented repository	A new GitHub project
Complete	The execution summary reached COMPLETE	Publishable without limitations
Rich / moderate / sparse	Presence of D, U and E artefacts under the fixed rubric	Accuracy, quality or reproducibility

Source: SOURCES.md; per-case execution_summary.json

Measurement scope

Measured 9 September 2026. File count and allocated size use regular files only and exclude .git, .DS_Store, caches and notebook checkpoints. Compressed files count once; bundled dependencies and retained outputs remain included. Locality describes retained analytical inputs, not software or API availability.

Source: VALIDATION_CORPUS.md; file_inventory.json

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Validation Corpus and Evidence

Pinned research repositories; each row is one unique repository

R1 — NYC-311

Anatomy: notebook-led · Evidence locality: external · Documentation: moderate

Pinned commit: ff013227b878e1960a48344534e560381938070b

Evidence basis: NYC311.ipynb; README.md:6; README.md:36; no input dataset in inventory; NYC311.ipynb: saved outputs; Project_Report_GIT.pdf

Open pinned repository

Source: validation_corpus.json; full inventory hashes in the evidence package.

R2 — World Bank poverty classification

Anatomy: notebook-led · Evidence locality: mixed · Documentation: rich

Pinned commit: d6c2ebf11fba9dc440b90d36240d5c6600b0ff00

Evidence basis: README.md: notebooks and execution order; notebooks/; README.md: Download the data; competition_winning_models_malawi/malawi/data/; README.md: Setup; requirements.txt; competition_winning_models_malawi/malawi/Bonus Prize/src/Full Bayesian Model Training and Predictions.ipynb

Open pinned repository

Source: validation_corpus.json; full inventory hashes in the evidence package.

R3 — GBD-Forecasting

Anatomy: research package · Evidence locality: mixed · Documentation: rich

Pinned commit: 3210f61191bff88cc3bcbdc6927327cfee065050

Evidence basis: pyproject.toml; src/__init__.py; src/pipeline.py; README.md: Data sources; data/modelling_dataset.csv; README.md: Reproducing the analysis; pyproject.toml; outputs/tables/full_model_comparison.csv

Open pinned repository

Source: validation_corpus.json; full inventory hashes in the evidence package.

R4 — Málaga parking digital twin

Anatomy: mixed-operational · Evidence locality: external · Documentation: moderate

Pinned commit: 3a7cfed3b045c41b7d938746849ccdda8c06a65c

Evidence basis: README.md: Architecture; kubernetes/; prediction-web/; spark-job/; README.md: Data Ingestion and Data Persistence; no parking input dataset in inventory; README.md: Quick Start; data-sink-job/requirements.txt; no retained project-specific analytical result located

Open pinned repository

Source: validation_corpus.json; full inventory hashes in the evidence package.

R5 — EPA CompTox-MIEML

Anatomy: notebook-led · Evidence locality: mixed · Documentation: moderate

Pinned commit: d15220ae813c8b77225e65b14adc91ad02f0f41c

Evidence basis: README.md: Running an Analysis; notebooks/ML_functions_vignette.Rmd; README.md: Additional Datasets; input/refchem.csv; input/LINCS_DTXSID_index_9_28_2020.csv; README.md: System Requirements; notebooks/ML_functions_vignette.nb.html

Open pinned repository

Source: validation_corpus.json; full inventory hashes in the evidence package.

Validation Corpus and Evidence

Pinned research repositories; each row is one unique repository

R6 — 19F NMR chemical shifts

Anatomy: notebook-led · Evidence locality: local · Documentation: moderate
Pinned commit: fe328d683287bca23e9058b514bff4049a15e324
Evidence basis: README.md: four notebooks and Recommended Execution Order; test.ipynb; README.md: Data Convention; data/raw/; data/processed/; README.md: Environment; requirements.txt; results/tables/; results/figures/

Open pinned repository

Source: validation_corpus.json; full inventory hashes in the evidence package.

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Validation Corpus and Evidence

Pinned research repositories; each row is one unique repository

R7 — LBNL city electricity use

Anatomy: notebook-led · Evidence locality: local · Documentation: moderate
Pinned commit: 6b69b255d309056edb7ad2360a48e4a027825d99
Evidence basis: README.md: Running; bin/section3.1 EDA.ipynb; README.md: Repository structure; data/; README.md: Set up the environment; requirements.txt; results/fig/; docs/paper_submitted.pdf

Open pinned repository

Source: validation_corpus.json; full inventory hashes in the evidence package.

R8 — EPA PFAS half-life

Anatomy: script-pipeline · Evidence locality: local · Documentation: moderate
Pinned commit: 0972b3defc02b8dbfdcfb3da7c7fcece4f7272cd
Evidence basis: README.md: Replication; CurrentScripts/; RData/Complete_Training_Data_JFW100322.csv; Predictors/; PFAS_HalfLifeData/; README.md: R version and Replication; Figures/; RData/

Open pinned repository

Source: validation_corpus.json; full inventory hashes in the evidence package.

R9 — Tornado prediction

Anatomy: script-pipeline · Evidence locality: external · Documentation: moderate
Pinned commit: 04ba066d0cbfdf12a4dee7d2543f1d640e2802d9
Evidence basis: README.md: General Information; preprocessing/; classification/; README.md: Preprocessing, ERA5/Storm Events and Google Drive datasets; no input datasets in inventory; README.md; images/ contains architecture/features illustrations; no retained fitted-model evaluation output established

Open pinned repository

Source: validation_corpus.json; full inventory hashes in the evidence package.

R10 — CDC influenza ferret model

Anatomy: script-pipeline · Evidence locality: mixed · Documentation: moderate
Pinned commit: debc57809def215c01ea8eb581af7ffc7dcdabc3
Evidence basis: README.md:26; DSU_working_forGitlhub.R (filename only); README.md:7 and 21–24; inputs/; README.md; inputs/FullModelMetricExample.csv; inputs/ModelMetrics4Heatmap.csv (presence only)

Open pinned repository

Source: validation_corpus.json; full inventory hashes in the evidence package.

R11 — Chelsea housing inspections

Anatomy: notebook-led · Evidence locality: not retained · Documentation: moderate
Pinned commit: 4bdc65d9b6c2d152a3e1fc2ba7d2191063ac52fd
Evidence basis: data-cleaning/data-cleaning.ipynb; models/; README.md:17–19; notebook file references; no input dataset in inventory; README.md; models/Predicting any code violation.ipynb; output/

Open pinned repository

Source: validation_corpus.json; full inventory hashes in the evidence package.

R12 — Insight Lane crash model

Anatomy: mixed-operational · Evidence locality: mixed · Documentation: rich
Pinned commit: 39b9c6958582f158adce8f746cd3844e3a29eec8
Evidence basis: Dockerfile; conf/; src/pipeline.py; src/models/train_model.py:59 and 133; README.md:82–86; src/data/tests/data/; src/data/weather/BostonWeather2016_Wunderground.csv; README.md: Setting up; requirements.txt; notebooks/general_analysis.ipynb

Open pinned repository

Source: validation_corpus.json; full inventory hashes in the evidence package.

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Validation Corpus and Evidence

Pinned research repositories; each row is one unique repository

R13 — SubwayScience

Anatomy: notebook-led · Evidence locality: mixed · Documentation: rich
Pinned commit: 85d08e9ad71367c1e5261f6bb756bf3b74dfc613
Evidence basis: TotalRidershipModels.ipynb; PerStationModels.ipynb; README.md; README.md:18–21; stationLocations.csv; README.md: Environment Setup; requirements.txt; plots/; TotalRidershipModels.ipynb

Open pinned repository

Source: validation_corpus.json; full inventory hashes in the evidence package.

R14 — Census BEACON

Anatomy: script-pipeline · Evidence locality: external · Documentation: moderate
Pinned commit: 133ae64c177e863bf1149872720cad01b0699346
Evidence basis: beacon.py; beacon_example.py; create_example_data.py; create_example_data.py:28–57; required NAICS XLSX files absent; README.md; beacon_example_output.txt; presentations/

Open pinned repository

Source: validation_corpus.json; full inventory hashes in the evidence package.

R15 — World Bank big-data poverty

Anatomy: script-pipeline · Evidence locality: external · Documentation: rich
Pinned commit: 7f5c1d31717e17026ee5efa9dd0fd852a5b87b70
Evidence basis: README.md:3–10; _main.R; DataWork/; README.md:14–28 and 55; no substantive input datasets retained; README.md: Setup and run code; renv.lock; Paper Tables and Figures/

Open pinned repository

Source: validation_corpus.json; full inventory hashes in the evidence package.

R16 — NOAA WoFS-ML-Severe

Anatomy: research package · Evidence locality: mixed · Documentation: rich
Pinned commit: 4e5087ec7d90e80856d38f09c412323d79d4faab
Evidence basis: setup.py: named packages; wofs_ml_severe/; README.md:32 and 45–51; tests/test_data/wofs_30M_07_20210504_2205_2235.nc; experiments/; README.md; setup.py; requirements.txt; images/performance_diagram.png; experiments/notebooks

Open pinned repository

Source: validation_corpus.json; full inventory hashes in the evidence package.

R17 — Cook County residential AVM

Anatomy: mixed-operational · Evidence locality: external · Documentation: moderate
Pinned commit: 4d52006653294b772a575c1f08fd1b029e2bd252
Evidence basis: README.md:106–138; dvc.yaml; pipeline/08-api.R; Dockerfile; README.md:1205–1211 and 1426–1437; .dvc/config; input/ has no retained observation dataset; README.md: Running and dependencies; renv.lock; reports/ contains templates rather than completed run results

Open pinned repository

Source: validation_corpus.json; full inventory hashes in the evidence package.

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Validation Corpus and Evidence

Pinned research repositories; each row is one unique repository

R18 — NASA ExoMiner

Anatomy: mixed-operational · Evidence locality: mixed · Documentation: rich
Pinned commit: 8960ddef576ab9474287630814bec9e2b9a596f6
Evidence basis: README.md:24–36; exominer_pipeline/; Dockerfile; exominer_vetting_pc_catalog_dash-render-web-app/; README.md:54–60; exominer_vetting_pc_catalog_dash-render-web-app/data/; README.md; docs/getting-started.md; docs/pipeline_outputs/; retained vetting catalogue

Open pinned repository

Source: validation_corpus.json; full inventory hashes in the evidence package.

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Validation Corpus and Evidence

Pinned research repositories; each row is one unique repository

R19 — CDC PPE concern detection

Anatomy: script-pipeline · Evidence locality: mixed · Documentation: rich
Pinned commit: d0932dd29e40a95f75b9f8b6a121c984e92ef96f
Evidence basis: README.md: Workflow; 01_clean-data/ through 05_results/; README.md:8–12; 03_ppe-coding/ml_dataset.csv; README.md: Setup Python environment; environment.yml; 05_results/results_ml.ipynb

Open pinned repository

Source: validation_corpus.json; full inventory hashes in the evidence package.

R20 — STARCOP

Anatomy: research package · Evidence locality: external · Documentation: rich
Pinned commit: c4789268a3fa0395f92357429052f6f5fc748acb
Evidence basis: setup.py; starcop/; scripts/train.py; README.md:42–49 and 84; no scene/label datasets in inventory; README.md: training; requirements.txt; notebooks/model_demos_AVIRIS.ipynb

Open pinned repository

Source: validation_corpus.json; full inventory hashes in the evidence package.

R21 — Regional HydroLSTM

Anatomy: script-pipeline · Evidence locality: mixed · Documentation: rich
Pinned commit: bb2696d4ed5555f6553aa9e77338b70b2aa1cb79
Evidence basis: README.md: Codes and Training; Code/main.py; Code/testing.py; README.md:26–43; Data/; Results/ absent; README.md: Getting started and Training; environment.yml; Notebook/Figures.ipynb

Open pinned repository

Source: validation_corpus.json; full inventory hashes in the evidence package.

R22 — GovTech KnowOrNot experiments

Anatomy: script-pipeline · Evidence locality: local · Documentation: moderate
Pinned commit: 86120ba67f36ca45017f98d6a0f0f190b6348549
Evidence basis: README.md: Usage and Repository Structure; pyproject.toml; experiment_run/; data/; questions_csv/; create_questions/create_ICA_questions.py:10; README.md: Usage; pyproject.toml; evaluation_results.csv; experiments/

Open pinned repository

Source: validation_corpus.json; full inventory hashes in the evidence package.

R23 — nhisml

Anatomy: research package · Evidence locality: external · Documentation: rich
Pinned commit: d287cb69450435c8693da6e5b1f4ba8fdba99c60
Evidence basis: pyproject.toml; src/nhisml/; tests/; README.md:96–110; src/nhisml/fetch.py; no NHIS microdata in inventory; README.md: CLI; pyproject.toml; docs/reference_statistics.md

Open pinned repository

Source: validation_corpus.json; full inventory hashes in the evidence package.

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Validation Corpus and Evidence

25 retained case records over 23 research repositories; the displayed cohort is a separate 23-case subset

Membership and outcome groups

Execution group	Case IDs	Repository relation	Outcome / inclusion
Research cases: publishable	R1, R2, R4, R8, R11, R12, R13, R19	Eight unique repositories	COMPLETE; PUBLISHABLE; included in displayed cohort
Research cases: publishable with limitations	R3, R5, R6, R7, R9, R14, R15, R16, R17, R18, R21, R22, R23	Thirteen unique repositories	COMPLETE; PUBLISHABLE_WITH_LIMITAT ONS; Q2_PARTIAL and S6_PROVISIONAL; included ir displayed cohort
Research attempts: not publishable	R10, R20	Two unique repositories	COMPLETE_WITH_FAILURES NOT_PUBLISHABLE; retained i R1–R23 corpus; excluded fror displayed cohort
Fresh rerun: NMR	R6_Fresh	Same repository family as R6	COMPLETE; PUBLISHABLE_WITH_LIMITAT ONS; included in displayed cohort
Fresh rerun: SubwayScience	R13_Fresh	Same repository family as R13	COMPLETE; PUBLISHABLE; included in displayed cohort

Source: manifest.json; execution_summary.json; execution_summary.json

Interpretation limit

These records establish artefact presence, execution state, stage coverage, publication eligibility and declared limitations. They do not establish independent replication, successful rerunning of repository code, model accuracy, external validity, or performance on 23 unseen projects. Snapshot equivalence between each Fresh copy and its earlier audit snapshot is not asserted here.

Source: VALIDATION_CORPUS.md; execution and source-copy manifests cited above

Signals Catalogue

Alpha boundary, identifiers and interpretation

Scope

This annex defines the 20 active deterministic checks presented in the TFM. TFM identifiers SG1–SG20 are consecutive presentation labels; Catalogue ID preserves the nonconsecutive native S5 identity. A Signal identifies an observable condition for review. It is neither a severity score nor a verdict on the project.

Source: Foundation S5 §§3–5; S5 Alpha Signal Catalogue v1.0-alpha

Four semantic outcomes

Outcome	Meaning
SIGNAL	Available evidence establishes the predefined review condition.
CLEAR	Available evidence positively supports the defined non-trigger condition.
INSUFFICIENT EVIDENCE	The required observations cannot support a deterministic decision.
NOT APPLICABLE	The check does not apply to the observed project context.

Source: Foundation S5 P3–P5; S5 Alpha Signal Catalogue activation summary

Failures are separate

Inspection or rule-execution failures are operational records isolated to the affected checks. They are not a fifth semantic outcome and must not be converted into CLEAR or SIGNAL.

Source: Foundation S5 P3 and operational contract

How to read the sources

Source keys state the conceptual basis or professional review rationale for each concern. They do not validate Bóveda's detector, implementation, thresholds or performance. Detector validation requires separate empirical testing, including positive, supported non-trigger and indeterminate cases.

Source: Foundation S5 P7, P9–P10

Signals Catalogue

Alpha boundary, identifiers and interpretation

Deferred candidates (not active checks)

Catalogue ID	Candidate	Reason deferred
SG7	Repeated/grouped entities cross partitions	Group semantics and the unseen-group claim are not deterministically available.
SG8	Training/evaluation samples overlap	Complete model-ready row identities across partitions are not exposed.
SG10	Duplicate model-ready observations	A complete normalized model-ready population is not exposed.
SG11	Predictors are not fewer than observations	Q1 has no typed N/P atoms, and S5 may not reinterpret compact LLM prose.
SG12	Class imbalance with naive evaluation	No defensible imbalance/metric threshold policy exists.
SG13	Missingness shift	No defensible materiality threshold exists.
SG14	Evaluation cohort changed by filtering	No defensible materiality policy exists.
SG17	Separate feature implementations diverge	Nominal cross-path feature correspondence requires unavailable semantics.
SG23	Meaningful simple baseline not found	No task-relative baseline taxonomy exists.
SG26	Data provenance/version not found	S2 lacks a normalized provenance-completeness state; parsing prose is prohibited.

Source: S5 Alpha Signal Catalogue, Deferred candidates

Signals Catalogue

TFM SG1–SG4

SG1 — Preprocessing fitted on evaluation data

Field	Definition
Identity	SG1 · Catalogue SG1 · SIG-EVAL-001
Family / type	Evaluation integrity · Defect-like
Required evidence	resolved train/evaluation roles, a state-learning operation, its fit input, and exact source locations.
Trigger	a verified preprocessing fit input includes the evaluation role.
Supported non-trigger	transform-only use, stateless transforms, or a training-only fit.
Limit / non-claim	conservative interprocedural flow may abstain; the Signal does not estimate bias magnitude or declare the model unacceptable.
Conceptual basis	KAU12, KAP23, SKL-PIT

Source: KAU12, KAP23, SKL-PIT

SG2 — Feature selection or dimensionality reduction fitted outside the training partition

Field	Definition
Identity	SG2 · Catalogue SG2 · SIG-EVAL-002
Family / type	Evaluation integrity · Defect-like
Required evidence	selector/reducer fit, partition operation, data role or source order, and exact locations.
Trigger	evaluation reaches the fit or the fitted operation demonstrably precedes the split.
Supported non-trigger	a selector/reducer fitted after splitting on training-only data.
Limit / non-claim	opaque pipelines and cross-file wrappers abstain; no performance effect is estimated.
Conceptual basis	SKL-PIT, KAP23

Source: SKL-PIT, KAP23

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Signals Catalogue

TFM SG1–SG4

SG3 — Resampling or class balancing applied before the evaluation split

Field	Definition
Identity	SG3 · Catalogue SG3 · SIG-EVAL-003
Family / type	Evaluation integrity · Defect-like
Required evidence	resampling operation, split, ordering or roles, and exact source locations.
Trigger	resampling includes evaluation-role data or demonstrably precedes the split.
Supported non-trigger	post-split training-only resampling.
Limit / non-claim	framework wrappers can leave order unresolved; class balance quality is not judged.
Conceptual basis	KAP23, SKL-PIT

Source: KAP23, SKL-PIT

SG4 — Final test/holdout used for model or hyperparameter selection

Field	Definition
Identity	SG4 · Catalogue SG4 · SIG-EVAL-004
Family / type	Evaluation integrity · Defect-like
Required evidence	evaluation role, metric/result, selection operation, and their data-flow connection.
Trigger	a resolved selection decision explicitly consumes final-evaluation information.
Supported non-trigger	selection driven by training/internal-validation evidence followed by separate final evaluation.
Limit / non-claim	only explicit data flow is active; opaque or indirect selection abstains.
Conceptual basis	VAR06, SKL-PIT

Source: VAR06, SKL-PIT

Signals Catalogue

TFM SG5–SG8

SG5 — Calibration or decision-threshold tuning reuses final evaluation data

Field	Definition
Identity	SG5 · Catalogue SG5 · SIG-EVAL-005
Family / type	Evaluation integrity · Defect-like
Required evidence	calibration/tuning operation, resolved data roles, final evaluation operation, and exact locations.
Trigger	the tuning fit explicitly includes evaluation-role data and evaluation is present.
Supported non-trigger	training/internal-calibration fitting with a separate final evaluation role.
Limit / non-claim	unresolved role aliases abstain; threshold adequacy is not judged.
Conceptual basis	SKL-CAL

Source: SKL-CAL

SG6 — Random or shuffled split used on a time-ordered forecasting path

Field	Definition
Identity	SG6 · Catalogue SG6 · SIG-EVAL-006
Family / type	Evaluation integrity · Review
Required evidence	temporal operation/path and split kind, both associated to the same source learning path, with exact locations.
Trigger	a temporal indicator and random/shuffled split are directly associated within one source function, notebook cell, or statement.
Supported non-trigger	an explicitly temporal split, or a project without established temporal applicability.
Limit / non-claim	technical temporal indicators do not prove user intent; cross-path or otherwise ambiguous association abstains.
Conceptual basis	SKL-TS, KAP23

Source: SKL-TS, KAP23

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Signals Catalogue

TFM SG5–SG8

SG7 — Target or direct target derivative included in model inputs

Field	Definition
Identity	SG7 · Catalogue SG9 · SIG-DATA-009
Family / type	Data, target, and representation integrity · Defect-like
Required evidence	supervised fit feature and target expressions with exact identities.
Trigger	the target literal/object is demonstrably present in the explicit feature input and not removed.
Supported non-trigger	an explicit feature contract that removes or separates the target.
Limit / non-claim	aliases and indirect derivatives abstain; the detector does not make semantic proxy judgments.
Conceptual basis	KAU12, KAP23

Source: KAU12, KAP23

SG8 — Training and inference feature schemas differ

Field	Definition
Identity	SG8 · Catalogue SG15 · SIG-CONSISTENCY-015
Family / type	Training / inference consistency · Defect-like
Required evidence	explicit train and inference feature names and exact source locations.
Trigger	both schemas resolve and differ.
Supported non-trigger	both explicit schemas match or a handled transformation establishes compatibility.
Limit / non-claim	dynamically generated schemas abstain; type compatibility is not inferred from names alone.
Conceptual basis	BRE17, ZIN-RML, AWS-FEAT

Source: BRE17, ZIN-RML, AWS-FEAT

Signals Catalogue

TFM SG9–SG12

SG9 — Training and inference preprocessing differ

Field	Definition
Identity	SG9 · Catalogue SG16 · SIG-CONSISTENCY-016
Family / type	Training / inference consistency · Defect-like / Review
Required evidence	both paths, role-resolved preprocessing operations, and exact locations.
Trigger	the resolved operation sets differ.
Supported non-trigger	identical explicit operation sets or one shared fitted pipeline.
Limit / non-claim	nominal correspondence and opaque orchestration abstain; semantic equivalence is not inferred.
Conceptual basis	SKL-PIT, ZIN-RML, BRE17

Source: SKL-PIT, ZIN-RML, BRE17

SG10 — Inference model differs from the model evidenced by evaluation

Field	Definition
Identity	SG10 · Catalogue SG18 · SIG-CONSISTENCY-018
Family / type	Training / inference consistency · Review / Defect-like
Required evidence	source-bound evaluation and inference model identities, plus explicit loaded-artifact identity where a mismatch is claimed.
Trigger	the evaluated and inference bindings resolve to different explicitly loaded model artifacts.
Supported non-trigger	the same source-bound model binding, or the same explicitly loaded model artifact, has both evaluation and inference evidence.
Limit / non-claim	metric namespaces, unrelated workflows, and model lineage beyond explicit source-bound identities abstain; equality of names across sources is not proof of binary identity.
Conceptual basis	BRE17, Professional ML review rationale (no external source asserted)

Source: BRE17, Professional ML review rationale (no external source asserted)

Signals Catalogue

TFM SG9–SG12

SG11 — Model artifact load lacks a resolved producing source reference

Field	Definition
Identity	SG11 · Catalogue SG19 · SIG-CONSISTENCY-019
Family / type	Training / inference consistency · Auditability / Review
Required evidence	literal model-load and model-save identities with exact locations.
Trigger	a loaded identity cannot be linked to any producing save identity.
Supported non-trigger	all model and possible-model loads have matching producing serialization references; same-source untyped pairs preserve this reference-level fact without establishing their model role.
Limit / non-claim	a SIGNAL requires positive model-role evidence. Unknown roles or dynamic identities abstain unless a producing reference is established for every possible-model load. CLEAR proves reference-level linkage only, not training history, execution, binary identity or contents. Missing in-boundary linkage does not prove lineage does not exist.
Conceptual basis	SCU15, BRE17

Source: SCU15, BRE17

SG12 — Stochastic data split lacks reproducibility control

Field	Definition
Identity	SG12 · Catalogue SG20 · SIG-REPRO-020
Family / type	Reproducibility and experimental discipline · Review
Required evidence	split API/kind, seed arguments or global seed operation, and exact locations.
Trigger	a stochastic split has no observed seed/random-state control.
Supported non-trigger	deterministic/temporal splits or explicit reproducibility control.
Limit / non-claim	unsupported framework controls can cause abstention; reproducibility does not imply methodological adequacy.
Conceptual basis	PIN21, NEU-CHK, AAAI-CHK

Source: PIN21, NEU-CHK, AAAI-CHK

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Signals Catalogue

TFM SG13–SG16

SG13 — Stochastic training lacks reproducibility control

Field	Definition
Identity	SG13 · Catalogue SG21 · SIG-REPRO-021
Family / type	Reproducibility and experimental discipline · Review
Required evidence	recognized model adapter, fit, seed/control, persistence operation, and exact locations.
Trigger	stochastic training lacks both reproducibility control and persisted exact state.
Supported non-trigger	explicit control or observable persisted exact model state.
Limit / non-claim	only documented framework adapters establish stochasticity; stable seeds do not guarantee full reproducibility.
Conceptual basis	PIN21, NEU-CHK, AAAI-CHK

Source: PIN21, NEU-CHK, AAAI-CHK

SG14 — Reproducible environment/dependency specification not found

Field	Definition
Identity	SG14 · Catalogue SG22 · SIG-REPRO-022
Family / type	Reproducibility and experimental discipline · Auditability
Required evidence	technical learning path plus deterministic project-file inventory.
Trigger	model fitting exists and no dependency specification is found.
Supported non-trigger	a recognized requirements, environment, lock, project, or container specification.
Limit / non-claim	"not found" is bounded to project evidence; file presence does not establish completeness.
Conceptual basis	PIN21, NEU-CHK, CODE-COMP

Source: PIN21, NEU-CHK, CODE-COMP

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Signals Catalogue

TFM SG13–SG16

SG15 — Single stochastic result reported without variability estimate

Field	Definition
Identity	SG15 · Catalogue SG24 · SIG-REPRO-024
Family / type	Reproducibility and experimental discipline · Review
Required evidence	stochastic model adapter, evaluation operation, variability/repetition operation, and exact locations.
Trigger	stochastic evaluation is observable and variability evidence is not found.
Supported non-trigger	repeated-run, bootstrap, confidence, error-bar, or dispersion evidence is observable.
Limit / non-claim	"not found" is not proof of absent reporting; unrecognized reporting mechanisms may be missed.
Conceptual basis	PIN21, NEU-CHK, AAAI-CHK

Source: PIN21, NEU-CHK, AAAI-CHK

SG16 — Core Data Shape incomplete

Field	Definition
Identity	SG16 · Catalogue SG25 · SIG-AUDIT-025
Family / type	Documentation and auditability · Auditability
Required evidence	validated Q1 structured state, record hash, and five field statuses.
Trigger	at least one fixed field is absent or unresolved.
Supported non-trigger	all five fields have a resolved structured status.
Limit / non-claim	S5 reads status fields only, never Q1 prose; "not found" does not prove nonexistence.
Conceptual basis	GEB21, NEU-CHK

Source: GEB21, NEU-CHK

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Signals Catalogue

TFM SG17–SG20

SG17 — Evaluation evidence not found

Field	Definition
Identity	SG17 · Catalogue SG27 · SIG-AUDIT-027
Family / type	Documentation and auditability · Auditability
Required evidence	fit and evaluation operations with exact locations and inspection coverage.
Trigger	fitting exists and evaluation evidence is not found.
Supported non-trigger	both fitting and a recognized evaluation operation are traceable.
Limit / non-claim	"not found" is bounded to supported evidence; it is not proof that evaluation never occurred.
Conceptual basis	MIT19, NEU-CHK

Source: MIT19, NEU-CHK

SG18 — Final or canonical model identity not found

Field	Definition
Identity	SG18 · Catalogue SG28 · SIG-AUDIT-028
Family / type	Documentation and auditability · Auditability
Required evidence	validated S4 structured canonical-status state and record hash.
Trigger	canonical status is explicitly unresolved.
Supported non-trigger	structured Yes or No with deterministically derived DOCUMENTED status.
Limit / non-claim	no S4 prose is interpreted; the Signal does not choose a primary model.
Conceptual basis	MIT19, Professional ML review rationale (no external source asserted)

Source: MIT19, Professional ML review rationale (no external source asserted)

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Signals Catalogue

TFM SG17–SG20

SG19 — Reproducible execution path not found

Field	Definition
Identity	SG19 · Catalogue SG29 · SIG-AUDIT-029
Family / type	Documentation and auditability · Auditability
Required evidence	learning operations plus deterministic entry-point inventory.
Trigger	learning exists and no execution entry point is found.
Supported non-trigger	an executable notebook, top-level workflow, language entry guard, Make/Snake/DVC workflow, or equivalent recognized entry file.
Limit / non-claim	"not found" is bounded; entry-point presence does not prove the full run will succeed.
Conceptual basis	PIN21, NEU-CHK, CODE-COMP

Source: PIN21, NEU-CHK, CODE-COMP

SG20 — Referenced data or intermediate artifact not found

Field	Definition
Identity	SG20 · Catalogue SG30 · SIG-AUDIT-030
Family / type	Documentation and auditability · Auditability
Required evidence	a literal in a recognized path-bearing read argument, resolution status, authorized boundary, and exact source location.
Trigger	an observed literal reference is MISSING or OUTSIDE_PROJECT.
Supported non-trigger	all observed literal references are present in-boundary or identify remote sources.
Limit / non-claim	dynamic/nested path construction and non-path API literals abstain; out-of-boundary means unavailable to this audit, not globally nonexistent.
Conceptual basis	GEB21, SCU15, NEU-CHK

Source: GEB21, SCU15, NEU-CHK

Technical Reproducibility and Access

Identify the delivery before reproducing it

Release identity

Item	Retained identity / interpretation
Hosted product	Bóveda Alpha 2.0.1 · https://boveda.dev
Source location	https://github.com/100471551/boveda-public-demo · Access is permission-dependent; a link alone does not establish public source availability.
UI release inspected	583f8d9f9b98beaa0c051cc0a8e253d48d1e7c41 · mobile audit-navigation revision.
Visual-evidence release	6961fb9322e3aa179328b5f40bb17c37cc94e4ab · expanded source-image gallery; canonical reports retained unchanged.
Analytical records	Preserved records have their own execution dates, configurations and evidence identities. A later UI commit is not a new analytical validation.
Thesis evidence package	TFM_Closure_2026-09-10 · corpus reconciliation, bibliography audit, R1 canonical record and provenance; see package manifest.

Source: Bóveda Alpha 2.0.1; product viewer and hosted-demo documentation

Three reproducibility questions

Inspecting the same preserved output, repeating the Bóveda reconstruction, and rerunning the source machine-learning project are different operations. The hosted demo supports the first. A new Bóveda run requires the preserved source boundary, analytical configuration, compatible environment and provider access. The Alpha does not arbitrarily execute the original project or promise to reproduce its reported metrics.

Source: Bóveda Alpha 2.0.1; preserved product implementation and release records

Credentials and access

Use the evaluator access supplied with the submission. Provider keys, deployment secrets and encryption keys are not part of this dossier or evidence package. Restricted repository or service access must be arranged separately; no public availability is implied.

Source: Bóveda Alpha 2.0.1; preserved product implementation and release records

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Technical Reproducibility and Access

Supported procedures and their limits

Procedure

Procedure	Prerequisites / command	What it establishes
Read the demo	Open boveda.dev; authenticate; open an available audit and its canonical report.	A consistent view of preserved outputs. The hosted runtime does not run the analytical pipeline.
Build hosted UI	Node 24; repository checkout. npm ci --ignore-scripts; npm run build:demo.	Builds public UI assets and validates bundle structure. Protected audit access also requires the documented server configuration.
Check hosted release	npm run demo:validate; npm run test:deploy.	Checks encrypted-bundle integrity and hosted behavior. Passing tests do not measure audit accuracy or reviewer agreement.
Open local application	From Boveda_2.0: python3 -B apps/primitive_probe/server.py --no-open --port 8724.	Starts the local viewer. Fresh execution requires the runtime freeze and provider credentials specified in the local setup.
Run a new audit	Choose a bounded project folder and an API spend guard; confirm Run audit locally.	Creates a new audit identity and output root. Budget checks occur after stages; a stage can exceed the guard.

Source: Bóveda Alpha 2.0.1; product viewer and hosted-demo documentation

Preservation boundary

The source selection is copied and verified by content before launch. The probe rejects symlinks, special files and unreadable subtrees rather than silently dropping them. A preserved source location is evidence of what the audit could inspect, not proof that external data or services remain accessible.

Source: Bóveda Alpha 2.0.1; preserved product implementation and release records

Known operational limits

The Alpha is not presented as a portable one-command reconstruction appliance. Native folder selection uses macOS; analytical execution follows a configured interpreter and retained freeze. Stronger worker isolation, durable recovery, cancellation and request-level resource controls remain hardening work.

Source: Bóveda Alpha 2.0.1; preserved product implementation and release records

Implementation sources

apps/primitive_probe/README.md; apps/primitive_probe/server.py; demo-v2/README.md; package.json; release records. These are implementation observations, not external validation results.

Source: Bóveda Alpha 2.0.1; preserved product implementation and release records

Technical Reproducibility and Access

Locate, verify and interpret the supporting artefacts

Evidence classes

Artefact	Establishes	Does not establish
Research corpus dataset	Pinned repositories, measured file scope and explicit anatomy/locality rubrics.	23 successful audits, model quality or complete data retention.
Execution / publication records	Attempt status, stage completion and publication eligibility.	Correctness of every reconstructed field.
Canonical report and evidence locators	The retained analytical claims and their traceable supporting locations.	Independent reproduction of the original project.
Test / verification summaries	Specified checks passed for an identified implementation and configuration.	Universal generalisation, false-positive rates or reviewer agreement.
Bibliographic source map	Conceptual foundations and guidance used in the dossier and Signal catalogue.	External endorsement or validation of Bóveda.

Source: Bóveda Alpha 2.0.1; product viewer and hosted-demo documentation

Historical verification example

The S5 R11–R16 validation summary records 120 scheduled and completed checks, no processing failures, 3,460 reverified evidence items and no snapshot mismatches. Its runtime and protocol commits are retained in that summary. This is a bounded historical test record, not a claim that every later release has the same empirical performance.

Source: outputs/S5_Signals/alpha_catalogue_validation_r11_r16_v1/validation_summary.json

R1 integrity example

The packaged canonical Markdown is accompanied by a SHA-256 provenance record. Annex E reproduces selected text without changing its field status, numerical value or claim boundary. Source paths and document hashes identify the retained record used for the example.

Source: Bóveda Alpha 2.0.1; preserved product implementation and release records

Submission handoff

Keep the supplementary manifest with the canonical record, corpus data and citation audit. File references in these annexes identify submission artefacts; they should travel with the final PDF. The web demonstration is a convenient inspection surface, not a substitute for the retained evidence package.

Source: Bóveda Alpha 2.0.1; preserved product implementation and release records

Representative Canonical Audit

R1 NYC-311: provenance and reading guide

Why R1

R1 is COMPLETE and PUBLISHABLE, all essential stages S1–S4 completed, and its package verification reports PASS with 587 checks and no errors. The assembled record preserves specialist sections, quantitative profiles, evidence identifiers and exact locators.

Source: execution_summary.json; package_verification.json; AUDIT_PUBLICATION_STATUS.json

Packaged evidence

Item	Role	Integrity
annex-e-r1-canonical-record.md	Full 484-line canonical record	cbdc9b5dd611bc660fa135d86d1b974ab88 43c95b8fd47ad49bdc0cf04c0096e
annex-e-r1-provenance.json	Source paths, hashes and scope	Machine-readable manifest
R1 stage records and evidence registries	Underlying validated records and locators	Referenced from the canonical record

Source: annex-e-r1-provenance.json

Reading rule

The canonical report states that it mechanically assembles validated specialist records and contains no Composer synthesis or cross-specialist reinterpretation. Status terms such as DOCUMENTED, INTERPRETED, DERIVED and UNRESOLVED must remain attached to their claims.

Source: S1_S6_Q1_Q2_Aggregate_Report.md

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Representative Canonical Audit

Exact text verified byte-for-byte against the packaged R1 record

Purpose

INTERPRETED — Characterize and predict patterns in NYC 311 service requests and their resolution. [E0001] [E0002] [E0004]

Source: S1_S6_Q1_Q2_Aggregate_Report.md — S1 Purpose → Core Purpose; E0001 README.md lines 1–51; E0002 NYC311.ipynb cells 1–20; E0004 NYC311.ipynb cells 71–130

Evidence basis

DOCUMENTED — The project actually relies on two external extracts: 2015 NYC 311 service-request records for the main analysis, and NOAA New York storm-event records for storm comparisons. [E0001] [E0004] [E0003]

Source: S1_S6_Q1_Q2_Aggregate_Report.md — S2 Evidence → Evidence Basis; E0001 README.md lines 1–51; E0004 NYC311.ipynb cells 1–25; E0003 NYC311.ipynb cells 71–130

Evidence gaps

UNRESOLVED — The exact source snapshots, extraction queries and storm-event coverage are not documented, preventing verification of completeness or version. The storm match ignores borough and lacks validation. The purported daily-volume series counts unnormalized Created Date timestamps: 1,403,536 distinct values are reduced to 391 timestamps having over 25 requests, so its interpretation as daily volume is unsupported. Representativeness of 311 reporting as a measure of underlying need is also unestablished. [E0001] [E0004] [E0003] [E0011]

Source: S1_S6_Q1_Q2_Aggregate_Report.md — S2 Evidence → Evidence Gaps; evidence locators preserved in the canonical record

Construction path

DOCUMENTED — The executed notebook reads 2015 NYC 311 requests, constructs a cleaned request table, then branches to: a Brooklyn HEAT/HOT WATER resolution-time classifier; an all-borough request-volume time series modeled with ARMA and Prophet; and a storm-linked volume series modeled with Prophet. [E0008] [E0009] [E0010] [E0011] [E0012] [E0013]

Source: S1_S6_Q1_Q2_Aggregate_Report.md — S3 Construction → Construction Path; NYC311.ipynb cells 1–135 as enumerated by E0008–E0013

Representative Canonical Audit

What this representative record supports

Supported and unsupported readings

Supported	Not established
A complete, publishable Bóveda audit artefact exists for R1	That the underlying NYC-311 project was independently rerun
Claims retain evidence IDs and source locators	That external datasets match a recoverable exact snapshot
The record distinguishes documented, interpreted, derived and unresolved claims	That all reported forecasts operate at a validated daily grain
The package passed its structural verification	That a single canonical final model was designated

Source: S1_S6_Q1_Q2_Aggregate_Report.md; package_verification.json

Material limitations to retain

The source snapshots and extraction queries are not documented; the storm match ignores borough and lacks validation; the purported daily-volume series groups unnormalized timestamps; split membership is not fixed by a seed; and multiple classifier/forecasting variants remain without one designated canonical final model.

Source: S1_S6_Q1_Q2_Aggregate_Report.md — S2 Evidence Gaps, S3 Construction Gaps, S4 Canonical Model Status

Representativeness boundary

R1 demonstrates the form and traceability of a canonical Bóveda output. It is illustrative rather than statistically representative, and it does not prove accuracy, external validity, reproducibility, or equivalent completeness across the corpus.

Source: annex-e-r1-provenance.json